Smart Space Optimisation at TU Berlin: Eliminating the Spatial Utilisation Gap

A Comprehensive Project Proposal & Business Plan (Full Central Library Edition)

Nasim Farhadi Balesht, Technische Universität Berlin • July 2026

1. Executive Summary: The Problem and The Solution

Technische Universität (TU) Berlin faces a dual challenge: meeting binding national climate protection laws while managing a historical campus footprint with a documented renovation backlog of approximately €2.4 billion. Under the European Green Deal and Germany's Federal Climate Change Act (*Klimaschutzgesetz*), public institutions are required to advance toward climate neutrality. TU Berlin has formally committed to cutting its direct greenhouse gas emissions by at least 470 tonnes by 2030.

While technical building retrofits have provided measurable improvements, a structural inefficiency remains: **The Spatial Utilisation Gap**. Because daily space occupancy is unpredictable, facilities management often expends excess energy heating, cooling, and lighting library wings and classrooms that are unoccupied.

The Solution: This proposal outlines the implementation of a decentralized, open-source Smart Desk-Utilization System. By deploying privacy-compliant radar sensors and an intelligent booking platform, the system delivers three primary outcomes:

1. **Efficient Building Utilisation:** It aligns mechanical heating, cooling, and lighting systems with actual occupancy, generating measurable financial and carbon emission reductions.
2. **Improved Resource Allocation:** It provides students with a reliable method to secure study spaces.
3. **Reduction of "Ghost Bookings":** It mitigates the campus-wide inefficiency of students blocking desks while remaining absent.

2. Strategic Scope & System Functionality

The project targets a full-scale rollout across the entirety of the TU Berlin Central Library, managing 1,243 quiet study desks across five levels.

The 1,243-Desk Target Zone:

1. **Ground Floor (Orange Zone):** Textbook Collection.
2. **1st Floor (Green Zone):** General interest and Computer Science.
3. **2nd Floor (Blue/Grey Zone):** Natural Sciences and Technology.
4. **3rd Floor (Purple Zone):** Humanities and Social Sciences.
5. **4th Floor (Yellow Zone):** Berlin University of the Arts (UdK) Library.

The "Waterfall" Logic: During off-peak hours, the system digitally locks the desk inventory on the upper levels, placing them into a low-energy "standby mode" where the Building Management System (BMS) reduces ventilation and heating. Students booking spaces are

naturally consolidated onto the Ground and 1st Floors. Once these lower floors reach an 85 percent capacity threshold, the system automatically unlocks Floor 2 and signals the BMS to initiate standard environmental controls, progressively unlocking higher floors only as human demand dictates.

3. Project Implementation Phasing

To mitigate technical and operational risks, the project is structured into two distinct implementation phases, ensuring all digital infrastructure is validated prior to physical deployment.

Phase 1: IT Infrastructure, Hardware Integration & Proof of Concept

Before modifying the library, this phase focuses entirely on technical development in a controlled environment. The core objectives are setting up the local open-source servers, establishing secure API bridges between the mmWave radar hardware and the central Building Management System (BMS), and finalizing the student booking application. Only upon successful integration testing will the project advance to physical installation.

Phase 2: The Full 5-Level Physical Rollout

Following Phase 1, the hardware will be deployed across the Central Library. This phase involves the electrical hardwiring of the ceiling sensors into the lighting grid, the application of desk QR codes, and the installation of the central floor visualization screens.

4. IT Infrastructure & The Open-Source Advantage

To avoid substantial recurring software licensing fees and maintain cybersecurity standards, the digital backbone relies on a decentralized, open-source architecture.

- **The Things Network (TTN):** The sensors transmit telemetry via The Things Network, an open wireless network operating on the license-free EU868 MHz band, routing data securely without cellular subscription costs.
- **Containerized Local Servers:** The software stack is hosted locally on a secure server managed by TU Campus IT.
- **Building Integration Bridge:** The open-source Data Hub translates occupancy data into standard building communication languages, integrating directly with the university's central heating and cooling controls.

5. Agile Hardware Architecture & Booking Mechanics

To respect student privacy and optimize installation logistics, the hardware utilizes a streamlined approach that minimizes physical modifications to the library furniture.

5.1 Multi-Zone mmWave Ceiling Radar

- **Technology:** The system utilizes commercial millimeter-wave (mmWave) radar sensors. Radar ensures GDPR compliance as it detects micro-movements (like breathing) but does not capture physical identities.
- **Sustainable Power Architecture:** To prevent the e-waste and maintenance burden of frequent battery replacements, the sensors will be hardwired into the existing overhead

lighting circuits using local low-voltage transformers. This provides a permanent power infrastructure with minimal installation disruption. Due to multi-zone tracking, approximately 310 nodes will sustainably cover the entire 1,243-desk footprint.

5.2 Desk Integration and Booking Workflows

The physical desks are managed through passive elements (QR code stickers) and centralized floor visualizations, supporting two clear booking workflows:

- **Advance Booking & Confirmation:** Students can reserve a desk in advance via the mobile app. Upon arrival at the library, they must confirm their booking by scanning either the physical QR code sticker at their designated desk, or the digital QR code integrated into the central floor screen at the entrance.
- **Spontaneous Direct Booking:** If a student arrives without a reservation, they can instantly claim any available space by directly scanning the physical QR code at the free desk or selecting it via the floor screen.
- **The Visual Protocol:** Both the central dashboards and the companion app display real-time status:
  - **Green:** Unreserved; the desk is available for direct booking.
  - **Blue:** Reserved, legitimately occupied, and verified by the ceiling radar.
  - **Red (Action Required):** The radar detects an individual at a desk without an active reservation, prompting the occupant to formalize their booking via the app.

6. Enhancing the Student Experience: Agile Development

A primary objective is optimizing how students navigate and share the library infrastructure. The exact adherence rules will be iteratively developed during the initial deployment phase, driven by direct student feedback and observed usage patterns.
Potential concepts to test during the pilot phase include:

- **Dynamic Grace Periods:** Implementing a 15-minute verification window to automatically release unattended desks and reduce "ghost bookings."
- **Peak Season Desk Lottery:** Randomly allocating highly contested time slots during examination weeks to ensure equitable access.
- **Soft Adherence Incentives:** Temporarily restricting advance booking privileges for users who consistently reserve spaces but fail to attend, encouraging fair spatial use among peers.

7. Business Case: Financials, Benchmarking & Partnerships

The operational model is streamlined to optimize the scale of a 5-floor deployment, ensuring long-term financial viability.

7.1 Strategic International IoT Partnerships

Hardware manufacturers actively seek practical testing environments to demonstrate their technologies within the European smart-building sector. In exchange for sponsoring the hardware footprint (ceiling radars and central floor dashboards), a partner receives a validated case study at TU Berlin.

- **The Open-Source Boundary:** To prevent vendor lock-in, the partner must supply hardware that transmits raw, openly readable data. TU Berlin retains full ownership of the resulting data and the open-source software stack.

7.2 CAPEX & OPEX Projections

By utilizing passive desk infrastructure (QR stickers) and leveraging the existing lighting grid for power, both capital and operational expenditures remain optimized. The installation budget accounts for certified electrical labor required across 5 floors.

CAPEX Category	Strategy	Projected Cost
Project Management	External agile project lead (12-month contract)	€45,000
Phase 1: IT Integration	External software contracting to build the app and link the open-source system with existing building climate controls	€25,000
Phase 2: Physical Installation	Electrical contractors for hardwiring 310 sensors into existing ceiling lighting circuits, plus 1,243 QR stickers and 5 floor dashboards	€65,000
Commercial Hardware	Absorbed via IoT Manufacturer sponsorship	€0
QA & Calibration	Radar sensitivity tuning and network testing	€15,000
Total CAPEX	**Comprehensive Full-Library Implementation Capital**	**€150,000**

OPEX Category	Description	Projected Annual Impact
System Support	System troubleshooting and logic tuning	- €15,000 / year
Utility Savings	Standby heating and cooling throttling across inactive floors during off-peak hours	+ €55,000 / year
Dynamic Housekeeping	Needs-based cleaning dispatches based on tracked desk usage across 1,243 spaces	+ €22,000 / year

Financial Rationale:
With an adjusted initial CAPEX of €150,000 and scaled net operational savings of €62,000 per year (Total Savings €77,000 minus €15,000 ongoing OPEX), the full-library rollout outlines a favorable direct financial payback period of **2.4 years**. Additionally, leveraging Open-Source software eliminates recurring proprietary licensing fees.

8. Architectural BIM Integration & Academic Handover

Beyond daily operational management, the Data Hub utilizes open APIs to export spatial occupancy data into TU Berlin's Building Information Modeling (BIM) infrastructure. This supplies university planners with empirical metrics to support precise Life Cycle Assessments (LCA).
To ensure institutional continuity, the open-source codebase and network framework will be integrated into the curricula of incoming TU Berlin computer science and engineering cohorts as an academic sandbox. Students will assist in repository maintenance as part of their coursework, fostering practical innovation while limiting long-term administrative costs.

References

1. Chalal, M. L., Medjdoub, B., Bezai, N., Bull, R., & Zune, M. (2022). Visualisation in energy eco-feedback systems: A systematic review of good practice. Renewable and Sustainable Energy Reviews, 162, 112447. https://doi.org/10.1016/j.rser.2022.112447
2. Delmas, M. A., Fischlein, M., & Asensio, O. I. (2013). Information strategies and energy conservation behavior: A meta-analysis of experimental studies from 1975 to 2012. Energy Policy, 61, 729–739. https://doi.org/10.1016/j.enpol.2013.05.109
3. European Commission (2019, updated 2021). The European Green Deal and the "Fit for 55" package: climate neutrality by 2050 and ≥ 55 % net emission cuts by 2030. https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en
4. Federal Republic of Germany (2021). Federal Climate Change Act (Bundes-Klimaschutzgesetz), 2021 amendment advancing climate neutrality to 2045. https://www.gesetze-im-internet.de/ksg/
5. Forschung & Lehre (2023). Report on the renovation backlog of German universities (TU Berlin ≈ €2.4 billion). https://www.forschung-und-lehre.de
6. Pritoni, M., Meier, A. K., Aragon, C., Perry, D., & Peffer, T. (2016). The iterative design of a university energy dashboard. ACEEE Summer Study on Energy Efficiency in Buildings. https://www.aceee.org/files/proceedings/2016/data/papers/8_466.pdf
7. Timm, S. N., & Deal, B. M. (2016). Effective or ephemeral? The role of energy information dashboards in changing occupant energy behaviors. Energy Research & Social Science, 19, 11–20. https://doi.org/10.1016/j.erss.2016.04.020
8. TU Berlin (2021). Klimaschutzvereinbarung 2021–2030. https://www.static.tu.berlin/fileadmin/www/10000000/Klimaschutz/KSV_TU_Berlin_2021-2030.pdf
9. TU Berlin / SDU (2023). Projekte zur Energieeinsparung und zum Klimaschutz 2023. https://www.static.tu.berlin/fileadmin/www/10004159/Managementsystem/Berichte_AUMS_ab_2022/2023_Energie_Klimaschutz_Projekte.pdf
10. Zangheri, P., Serrenho, T., & Bertoldi, P. (2019). Energy savings from feedback systems: A meta-studies' review. Energies, 12(19), 3788. https://doi.org/10.3390/en12193788
11. Additional dashboards consulted: Cornell University, sustainable.cornell.edu/buildings-energy/building-energy-dashboard; University of British Columbia, facilities.ubc.ca; Stony Brook University, stonybrook.edu/energy/dashboard; Oberlin College; Brock University; University of Connecticut; Keio University.